EEE 6778. Applied Machine Learning II

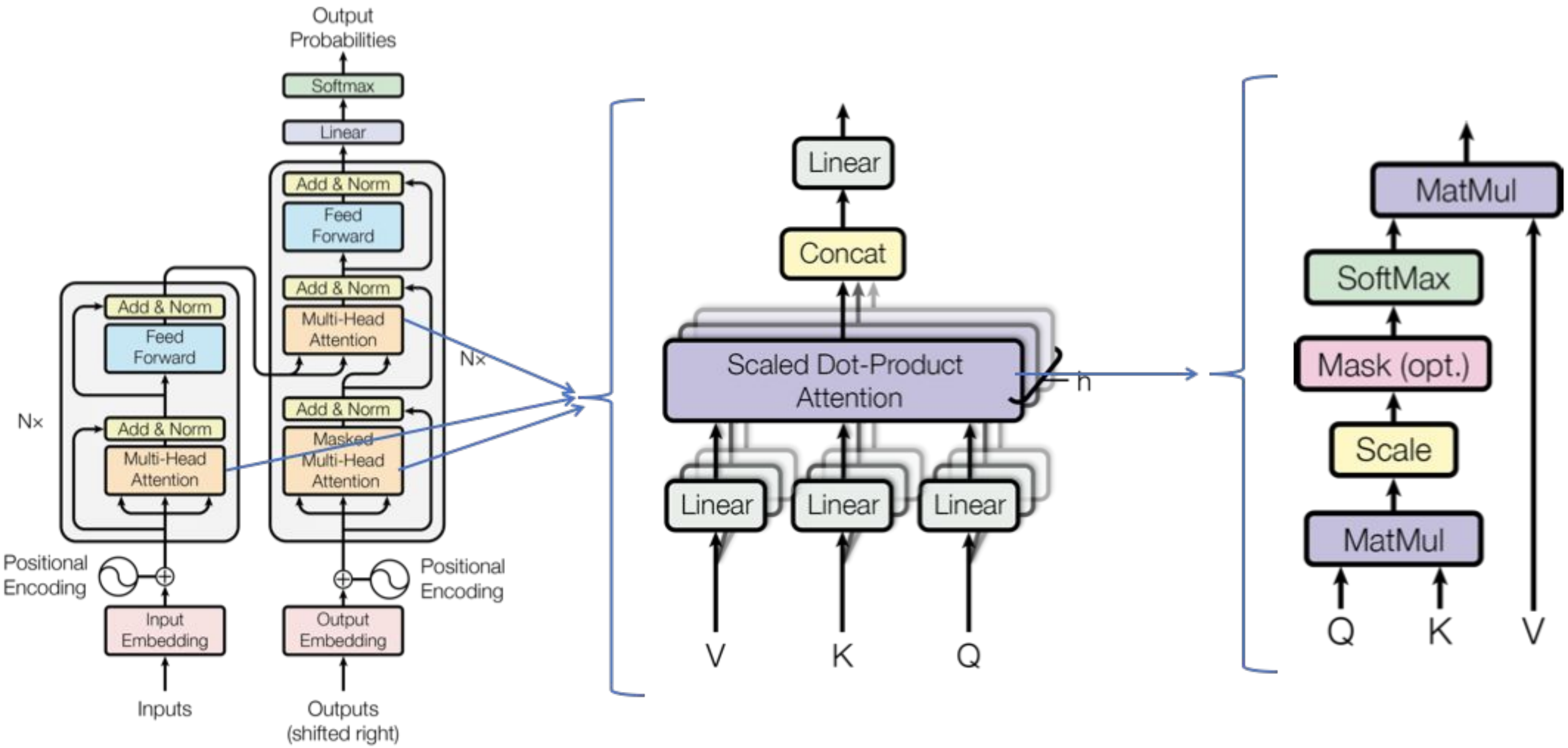
UNIVERSITY of FLORIDA

Module 5. Transformers

Transformers Language Models
Training in Transformers
Multimodal Transformers

Transformers Recap

From Multi-Head Attention to Transformers



https://medium.com/towards-data-science/self-attention-and-transformers-882e9de5edda

Encoder Transformer - (Understanding)

Each encoder layer consists of:

- Multi-Head Self-Attention (each word attends to all words)

$$Z = \mathrm{MultiHead}(X)$$

- Residual Connections + Layer Normalization

$$Z_{\mathrm{norm}} = \mathrm{LayerNorm}(X + Z)$$

- Feedforward Network (FFN)

$$F = \max(0, Z_{\mathrm{norm}} W_1 + b_1) W_2 + b_2$$

- Final Layer Normalization (Output is an embedding)

$$Y = \mathrm{LayerNorm}(Z_{\mathrm{norm}} + F)$$

Decoder Transformer - (Generation)

GPT, LLaMA, Mistral → Used for text generation, chatbots, story writing.

The decoder is similar to the encoder but with two major differences:

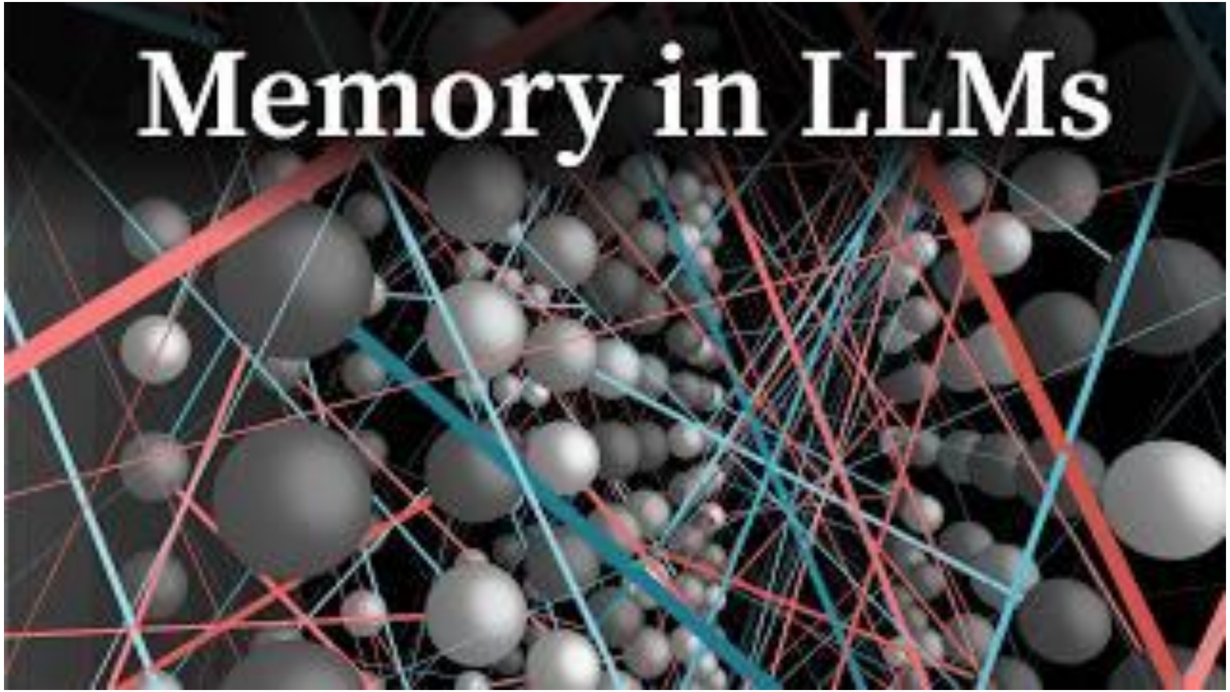
- Masked Self-Attention
    - Prevents looking at future tokens to ensure autoregressive generation.
    - Uses a triangular mask so token t_i only sees previous tokens.
- Cross-Attention to Encoder
    - The decoder gets context from the encoder.
    - It uses encoder outputs as Keys and Values but keeps Queries from the decoder.

Instead of using self-attention, we now use the encoder outputs as context

$$\mathbf{A} = \frac{\mathbf{Q}_{\text{decoder}}\mathbf{K}_{\text{encoder}}^T}{\sqrt{d_k}}$$

$$\mathbf{Z} = \text{softmax}(\mathbf{A})\mathbf{V}_{\text{encoder}}$$

From Multi-Head Attention to Transformers



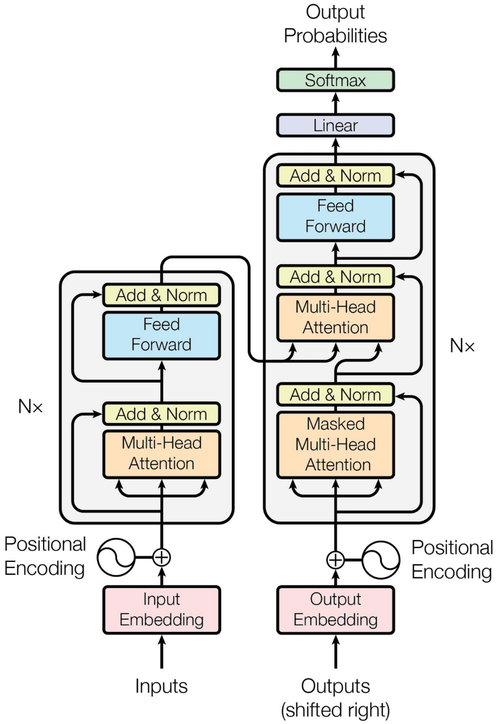
Source:
3Blue1Brown

Encouragement to Explore

Transformers in PyTorch

Training in Transformers

Transformers Architecture



Causal Language Modeling (CLM) – GPT, LLaMA, Mistral

- **How it Works:** Predict the **next token** given past tokens (left-to-right).
- **Masking**: Causal masking (prevents peeking at future words).
- **Ground Truth:** The actual next token in a dataset.
- **Loss Function:** Cross-entropy loss on next-word prediction.
- **Examples**: GPT-3, GPT-4, LLaMA, Mistral, Falcon.
- **Use Cases:** Text generation, chatbots, code completion.

Input	Target (Next Token)
"The cat sat"	"on"
"The cat sat on"	"the"

Reinforcement Learning (RLHF) – ChatGPT, Claude, Gemini

- **How it Works:** A base LLM is fine-tuned using human feedback to improve response quality.
- **Masking**: No explicit token prediction; trained to maximize rewards.
- **Ground Truth:** Human preference rankings of responses.
- **Loss Function:** Reward maximization (PPO in RLHF).
- **Examples:** ChatGPT (GPT-4-turbo), Claude, Gemini.
- **Use Cases:** Fine-tuning AI chatbots, aligning AI with human intent.
- Example Training Process
    - Model generates multiple responses to a prompt.
    - Human labelers rank responses from best to worst.
    - The model learns from the preferred responses.
    - A reward model is trained, and Proximal Policy Optimization (PPO) updates the LLM.

Masked Language Modeling (MLM) – BERT, RoBERTa, DeBERTaMasked Language Modeling (MLM) – BERT, RoBERTa, DeBERTa

- **How it Works:** Some words in the sentence are masked, and the model predicts them.
- **Masking**: Random masking (hides tokens in input).
- **Ground Truth:** The original token before masking.
- **Loss Function:** Cross-entropy loss on masked token prediction.
- **Examples**: BERT, RoBERTa, DeBERTa, ALBERT.
- **Use Cases:** Text embeddings, classification, question answering.

Input	Target (Masked Token)
"The cat [MASK] on the mat."	"sat"

Implementation

Scratch version of a Transformer implementation is on the repository!

Analyze the outputs to understand the process

Multimodal Transformers

Text Generation - LLM

- **The Final Step in the Decoder Transform Architecture was:** A softmax layer selects the most probable next word.
- LLM models generate content based on probabilities

Beyond Text

- Transitioning from LLMs to multimodal systems involves extending the core transformer principles—**attention, multi-head mechanisms, and sequential processing—to additional modalities.**
- By incorporating modality-specific encoders, employing effective fusion strategies, and leveraging cross-modal attention, you can build systems that not only understand text but also integrate and generate insights from **images, audio, and more.**

How?

When transitioning from LLMs to multimodal systems, one of the first key steps is to design encoders that can extract rich, meaningful features from non-textual data

Images

- CNN
- ViT

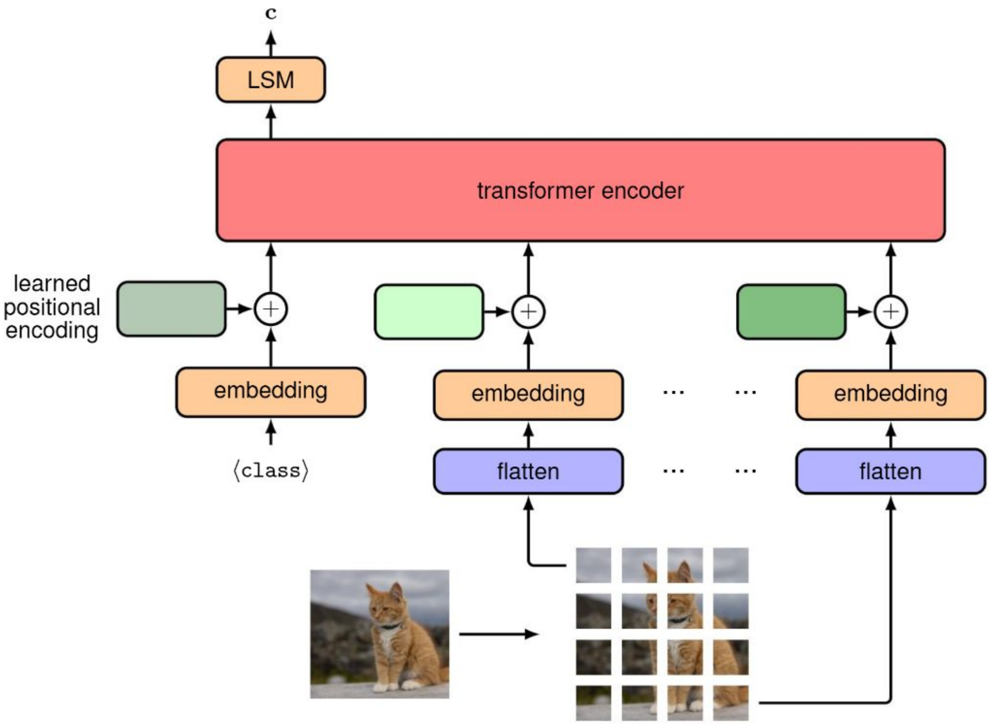
Audio

- RNN and LSTM
- Audio Transformers

Video

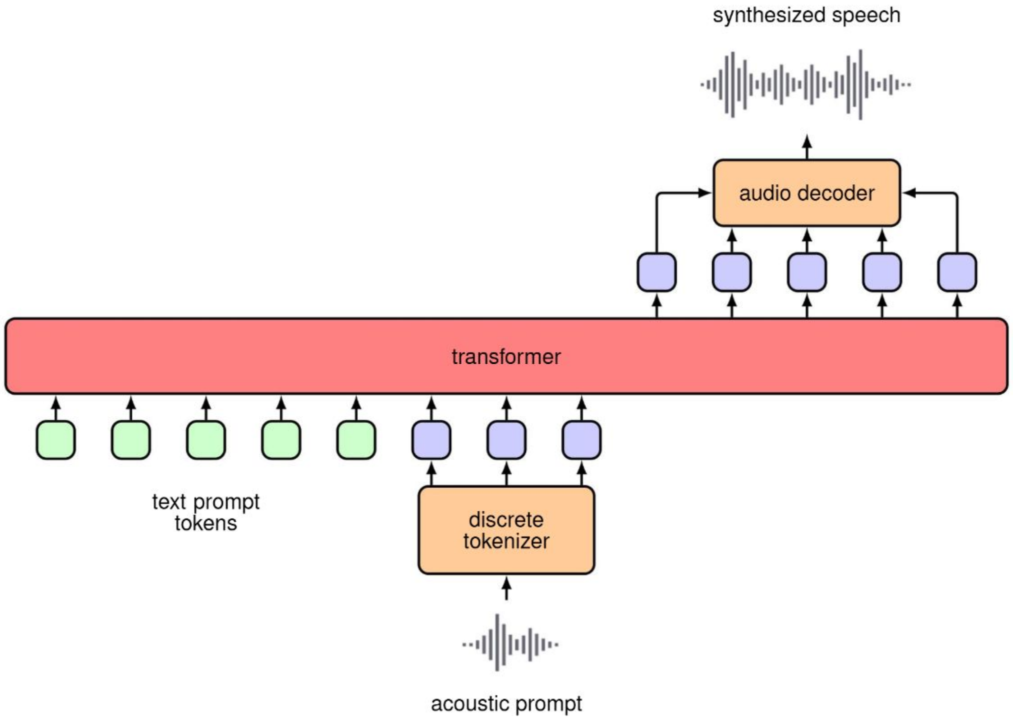
- 3D CNNs
- Video Transformers

Vision Transformers



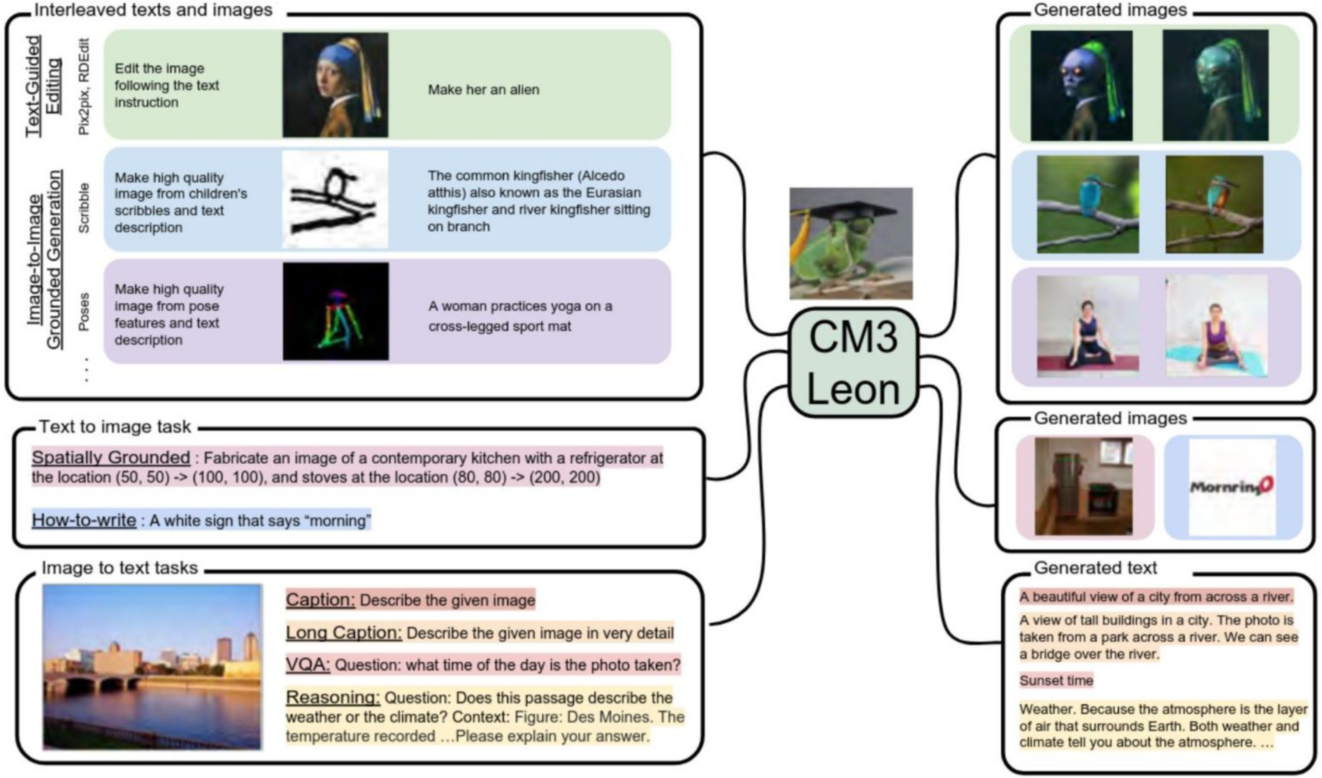
(Dosovitskiy, 2020)

Audio Transformers



(Wang et al., 2023)

Vision and Language Transformers



(Yu et al., 2023)